LAURA CONNOLLY

Email: laura.connolly@queensu.ca
LinkedIn: `https://www.linkedin.com/in/laura-connolly-0aab43144`

CURRENT POSITION

Malone / NSERC Postdoctoral Fellow *Sept. 2025 - present*
Malone Center for Engineering in Healthcare in the Whiting School of Engineering at Johns Hopkins University, Baltimore, USA

- Supervised by Professors Russell H. Taylor and Axel Krieger
- Working on autonomous medical robotics and robot-assisted nephrectomy. Press release available here: malone-postdoctoral-fellow-announcement

EDUCATION

Doctor of Philosophy (Ph.D.) *2020 - 2025*
Queen's University, Kingston, Canada
Electrical Engineering

- Mentors: Dr. Gabor Fichtinger (Queen's), Dr. Parvin Mousavi (Queen's), and Dr. Russell H. Taylor (Hopkins)
- Promoted to Ph.D. without completion of Master's degree
- NSERC CREATE Training Program in Medical Informatics and Vector Institute – Trainee
- Collaborative Biomedical Engineering (CBME) – Specialization

Bachelor's of Applied Science (B.A.Sc.) *2016 - 2020*
Queen's University, Kingston, Canada
Electrical Engineering

- Graduated with First Class Honours
- Biomedical Engineering Specialization

Relevant Coursework:

- **Fall School on Medical Robotics** *2021*
  Georgia Institute of Technology (Georgia Tech), United States (In-person)
- **Summer School on Image Processing** *2020*
  University of Szeged, Hungary (Online)
- **Hamlyn Winter School on Surgical Imaging and Vision** *2020*
  Imperial College London, United Kingdom (Online)

AWARDS AND SCHOLARSHIPS

Scholarships & grants

- Malone Postdoctoral Fellowship - Johns Hopkins Malone Center for Engineering in Healthcare *2025*
- NSERC Postdoctoral Fellowship (PDF) *2025*
- Vector Research Grant *2023-2025*
- NSERC Michael Smith Foreign Study Supplement (MS-FSS) *2023*
- Queen's University Tri-Agency Recipient Recognition Award (TARRA) *2022-2023*
- Walter C. Sumner Memorial Fellowship and Renewal *2022-2023*
- NSERC Canada Graduate Scholarship - Doctoral (CGS-D) *2022-2025*
- MediCREATE Training Award – NSERC CREATE Training Program in Medical Informatics *2021*
- Mitacs Globalink Research Award *2020 - deferred to 2021*
- NSERC Alexander Graham Bell, Canada Graduate Scholarship-Master's (CGS-M) *2020-2021*
- NSERC Undergraduate Student Research Award (USRA) *2019 & 2020*

Research awards

- Best Application Award - Hamlyn Surgical Robotics Challenge *2025*
- Runner-Up Award for Siemens Healthineers Best Paper IPCAI - Co-author on [J-4] *2025*
- Runner-Up - IEEE Research Excellence Award - Ph.D. category - Kingston Section *2023*

- Best Demo Award - Advances in Simplifying Medical UltraSound (ASMUS) - MICCAI Workshop *2023*
- Best Student Paper Award – IEEE International Conferences on Autonomous Systems (ICAS) *2021*
- Student's Choice Award – First Place – Electrical and Computer Engineering Capstone Demo Day *2020*
- PEO Student Paper Competition Finalist *2019*

Presentation awards

- IPCAI ImFusion Long Abstract Audience Choice Award *2025*
- IPCAI J&J MedTech Best Presentation Audience Choice Award *2025*
- Best Presentation Award (Image-guided Interventions and Surgery) - ImNO x IGT Symposium *2025*
- Best Presentation Award (Image-guided Interventions) - Imaging Network of Ontario (ImNO) *2024*
- Runner-Up - Queen's 3 Minute Thesis Competition *2021*
- Best Pitch Award (Image-guided Surgery) – Imaging Network of Ontario (ImNO) *2021*

Reviewer awards

- Outstanding Reviewer Award – IPCAI (selected out of 147 reviewers) *2024*
- Outstanding Reviewer Award – MICCAI (selected out of 1659 reviewers) *2023*

Travel awards

- IPCAI Student Participation Award *2025*
- SPIE Medical Imaging - Travel Bursary *2022 - deferred*
- SPIE Medical Imaging - Travel Bursary *2020*

RESEARCH AND PROFESSIONAL EXPERIENCE

Visiting Graduate Scholar *Nov 2021 - Apr 2022 / Mar 2023 - Aug 2023*
Laboratory for Computational Sensing and Robotics (LCSR), Johns Hopkins University, Baltimore, USA

- Supervised by Professors Russell H. Taylor and Simon Leonard
- Built and deployed an open-source platform for image-guided robotics research: `https://github.com/rosmed/slicer_ros2_module`
- Research culminated in 3 journal papers (1 under review), 1 conference paper, 4 international workshop presentations, and a provisional patent application

Medical Systems Engineering Consultant *May 2020 - May 2025*
Medical Robotics Research Group

- Technical consultant and developer for a surgical navigation prototype

Student Researcher *May 2018 - July 2025*
Laboratory for Percutaneous Surgery (Perk Lab), Queen's University
Medical Informatics Lab (Med-i), Queen's University

- Summer 2018: Reprogrammed a 3D printer to scan tissue samples for data collection
- Summer 2019: Developed, trained, and tested a classifier to distinguish cancer tissue using mass spectrometry
- Graduate studies: Research on robot-assisted breast cancer surgery

Lab Technician *Jan 2020 - May 2020*
BioRobotics Research Laboratory, Queen's University, Kingston

- Responsible for running tests and supervising the 6 DOF Quanser/ CRS Model A465 Open Architecture Robot for graduate students in ELEC 848: Control System Design for Robots and Telerobots

Equity, Diversity, Inclusion, and Indigenization (EDII) Officer *Nov 2023 - August 2025*
Perk Lab, Queen's University

- Responsible for organizing seminars related to EDII topics and creating the first lab code of conduct

Technical Lab Coordinator *Jan 2021 - Sept 2023*
Perk Lab, Queen's University

- Technical lab coordinator for the Perk Lab – a research group of approximately 30 undergraduates, engineers, post-doctoral fellows, and graduate students - responsible for acquisitions, expense and resource management

Chapter Leader – Makers Making Change Kingston *Jan 2021 – Nov 2021*
A Neil Squire Program

- Chapter leader for the Queen's University chapter of Makers Making Change (MMC), a non-for-profit that 3D prints devices for accessibility

Investment Associate *May 2019 - July 2020*
Front Row Ventures

- Front Row Ventures is the first student-run venture capital fund entirely managed by students in Canada - The investment team is responsible for meeting with student founders and allocating $25000 investments

Engineering Society Involvement
Queen's University

- Faculty of Engineering and Applied Science Orientation Leader *Aug 2017 – Sept 2017*
- Speaker's Coordinator - First Year Conference *Nov 2017 – Jan 2018*
- Integrated Learning Center Student Constables (iCon) - Lab Supervisor *Sept 2017 – June 2019*

TEACHING EXPERIENCE

Teaching Assistantships
Queen's University

- Graduate TA - Artificial Intelligence *Jan 2025 - March 2025*
- Graduate TA - Introduction to Robotics *Jan 2024 - May 2024*
- Graduate TA - Electrical and Computer Engineering Design and Practice *Sept 2023 - Dec 2023*
- Graduate TA - Electrical and Computer Engineering Design Project (Capstone) *Sept 2020 - May 2023*
- Project Manager - Engineering Design and Practice Sequence (Module 3) *Sept 2019 – Apr 2020*

Healing Through Collaboration: Open-Source Software in Surgical, Biomedical, and AI Technologies
Hamlyn Symposium for Medical Robotics *2025*

- As a co-organizer of this workshop at HSMR 2025, I helped with speaker coordination, poster review, general planning, and building the website. More details are available here: healing-through-collaboration

Workshops on Open-Source Software for Image-Guided Robotics
International Symposium for Medical Robotics *2023-2025*

- As a member of the ROS-MED research team, I have helped organize and present three workshops to teach members of the community how to build open-source image-guided robotic systems (event-history)

Electrifying Medicine
Queen's University *April 2024*

- Organized a hands on session for 30 local grade six students on robotics and the basics of electricity

MD Meets Machine
Queen's University *July 2022 - Aug 2022*

- Helped run and prepare a workshop for high-school students interested in medical image computing research
- More details on this event are available here: news-article

MENTORSHIP EXPERIENCE

As a senior graduate student in the Perk Lab and Med-i Lab at Queen's University, I had the opportunity to mentor several undergraduate and graduate students. Below is a list of projects on which I have served as a mentor, along with some notable mentorship achievements:

1. **Spectroscopic tissue scanning** (Sept. 2021 – Sept. 2023)
   I mentored an M.A.Sc. student who continued my work (presented in C-10) on tissue classification using spectroscopy. Together, we explored techniques for tracked scanning of ex vivo tissue and machine learning-based spectral classification. This collaboration resulted in both a conference (C-9) and a journal publication (J-9).

2. **Tracked ultrasound imaging for tumor bed inspection** (May 2022 - Aug. 2022)
   I mentored an undergraduate Biomedical Computing student through the Google ExploreCSR program (news-article). We developed a prototype navigation system for ultrasound-guided tumor bed inspection, which she presented as an oral presentation at the SPIE Medical Imaging Conference in 2023 (C-8). She is now a Master's student and I continue to collaborate with her on mass spectrometry tissue imaging (J-3).

3. **3D-printed prosthetics for Burma Children's Medical Fund** (May 2023 – *present*)
I helped initiate a collaboration with the BCMF clinic in Thailand, which serves refugees of the Burma civil war by providing 3D-printed prosthetics to patients lacking conventional healthcare access. We have since sent two students to the clinic, whom I trained in 3D printing and prototyping and supported throughout their time abroad. I have also contributed to several abstract submissions related to this work (A-5, A-10, A-12, A-15). This work received third place at the Health and Human Rights Conference at Queen's University, two prizes totaling $4,000 USD at the Rice360 Global Health Technologies Design Competition, and first place overall ($1000 USD) at the RESNA conference in Chicago.

4. **Robotic tumor bed inspection** (May 2024 – Aug. 2024)
I was awarded a Med-i CREATE Summer Research Award to support an undergraduate mechatronics student in summer 2024. We were also joined by a student from the Engineering Science program at the University of Toronto. Together, we developed a low-cost bench-top system for tumor bed inspection using a 6-axis robot. Each student had a first-author paper accepted and presented at the SPIE Medical Imaging Conference (C-3, C-4), and one student won third place in the JMI pitch competition.

5. **Autonomous ultrasound scanning** (Sept. 2024 – March 2025)
I mentored two projects (Grade 5 & 7) for the Frontenac, Lennox, and Addington Science Fair. Both projects won special awards and one won a silver medal in the engineering category. I also served as a category judge.

6. **Robotic skin cancer detection** (Sept. 2024 – *present*)
I am currently mentoring an M.A.Sc. student in the Perk / Med-i Lab who is optimizing the robotic scanner we developed in C-4 for skin cancer detection. We meet regularly and are preparing a conference submission to be finalized in fall 2025.

7. **Wearable haptic feedback for tumor localization** (May 2025 – *present*)
I secured funding to support another undergraduate mechatronics student developing a wearable haptic feedback bracelet for tumor localization in breast surgery. We just submitted his work to SPIE Medical Imaging 2026 (C-1).

SELECTED WORKSHOPS AND INVITED TALKS

Please note that some of the presentation titles are hyperlinked to articles or recordings of the presentation.

Guest Lecturer - CIS I: Computer-Integrated Surgery — *Upcoming - October 2025*
Johns Hopkins University

Virtual Open Forum on Computer- and Robotics-Assisted Cancer Surgery — ***September 2025***
Machine Learning in Medical Imaging Consortium (MaLMIC)

Rising Star Talk — ***May 2025***
ICRA Robot-Assisted Medical Imaging Workshop (RAMI)

Keynote Speaker - STEM Horizons — ***March 2025***
AI & Healthcare Summit

Debate Participant: The Role of AI in Medicine — ***March 2024***
Imaging Network of Ontario Symposium / NSERC CREATE Program for Medical Informatics and Responsible AI

Guest Lecturer - CISC 330: Computer Integrated Surgery — ***Nov 23 2023***
Queen's University, School of Computing

Invited Speaker: Open-source software for surgical technologies workshop — ***June 2023***
Hamlyn Symposium on Medical Robotics (HSMR), London, UK

Presenter: Building Software Systems for Image-Guided Robot-Assisted Interventions — ***April 2023***
International Symposium on Medical Robotics (ISMR), Atlanta, USA

Invited Panelist — ***March 2021***
Queen's Women in Engineering and Applied Science Conference (Q-WASE)

ACADEMIC SERVICE

- **Reviewer for:** IEEE Transactions on Biomedical Engineering, IEEE Robotics and Automation Letters, IEEE International Conference on Intelligent Robots and Systems (IROS), SPIE Journal of Electronic Imaging, SPIE Medical Imaging Conference - Sub-reviewer, Medical Image Computing and Computer Aided Intervention Conference (MICCAI), International Conference on Information Processing in Computer-Assisted Interventions (IPCAI), The Hamlyn Symposium on Medical Robotics (HSMR) - Sub-reviewer, IEEE International Conference on Robotics and Automation (ICRA)
- **Scholarship Committees:** Vector Scholarship in AI, NSERC Scholarships and Fellowships Selection Committee for Computing Sciences (198)
- **Area Chair:** IPCAI 2026
- **Student Representative:** Med-i CREATE Program Committee, 2022 - 2025

PATENTS

Photoacoustic tumor bed imaging in surgery
Provisional patent filed Oct 2023

- Inventors: Russell H. Taylor, Emad Boctor, **Laura Connolly**

PUBLICATIONS

Google Scholar: `https://scholar.google.ca/citations?user=9E-xfIwAAAAJ&hl=en`
Journal Articles

J-1 **L. Connolly**, C. Yeung, H. Ishida, T. Ungi, A. Munawar, A. Deguet, R.H. Taylor, G. Fichtinger, P. Mousavi, K. Hashtrudi-Zaad. Are you certain you want to cut there? Uncertainty-based virtual fixtures for breast-conserving surgery. IEEE Robotics and Automation Letters (In review).

J-2 **L. Connolly**, H. Song, K. Xu, A. Deguet, S. Leonard, G. Fichtinger, P. Mousavi, R.H. Taylor, E. Boctor. No cancer left behind: A testbed and proof of concept for photoacoustic tumor bed inspection. The Journal of Computer Assisted Surgery, Taylor & Francis (Under revision).

J-3 O. Radcliffe, **L. Connolly**, A. Jamzad, M. Kaufmann, S. Merchant, J. Engel, R. Walker, S. Varma, G. Fichtinger, J. Rudan, and P. Mousavi. Anomaly Detection using Intraoperative iKnife Data: A Comparative Analysis in Breast Cancer Surgery. Accepted to the International Journal of Computer Assisted Radiology and Surgery (IJCARS), 2025 (https://doi.org/10.1007/s11548-025-03476-0).

J-4 **L. Connolly**, T. Ungi, A. Munawar, A. Deguet, C. Yeung, R.H. Taylor, P. Mousavi, G. Fichtinger, K. Hashtrudi-Zaad. Touching the Tumor Boundary: A Pilot Study on Ultrasound-Based Virtual Fixtures for Breast-Conserving Surgery. International Journal of Computer Assisted Radiology and Surgery, special issue of International Conference on Information Processing in Computer-Assisted Interventions (IPCAI), 2025 (https://doi.org/10.1007/s11548-025-03342-z). (Shortlisted for Best Paper Award at IPCAI 2025).

J-5 M. Farahmand, A. Jamzad, F. Fooladgar, **L. Connolly**, M. Kaufmann, K. Yi Mi Ren, J. Rudan, D. McKay, G. Fichtinger, P. Mousavi. FACT: Foundation Model for Assessing Cancer Tissue Margins with Mass Spectrometry.International Journal of Computer Assisted Radiology and Surgery, special issue of International Conference on Information Processing in Computer-Assisted Interventions (IPCAI), 2025 (https://doi.org/10.1007/s11548-025-03355-8)(Runner-Up Award for Siemens Healthineers Best Paper Award - IPCAI 2025).

J-6 M. Sahu, H, Ishida*, **L. Connolly***, H. Fan*, A. Deguet, P. Kazanzides, F.X. Creighton, R.H. Taylor, A. Munawar. Integrating 3D Slicer with a Dynamic Simulator for Situational Aware Robotic Interventions. IEEE Transactions on Medical Robotics and Bionics (T-MRB), 2024. 6(4), 1405-1408. (doi.org/10.1109/TMRB.2024.3475827). * denotes equal contribution

J-7 **L. Connolly**, A. Kumar, K.K. Mehta, L. Zogbi, P. Kazanzides, P. Mousavi, G. Fichtinger, A. Krieger, J. Tokuda, R.H. Taylor, Simon Leonard, A. Deguet. SlicerROS2: A research and development module for image-guided robotic interventions. IEEE Transactions on Medical Robotics and Bionics (T-MRB), 2024. 6(4), 1334-1344. (doi.org/10.1109/TMRB.2024.3464683).

J-8 **L. Connolly***, F. Fooladgar*, A. Jamzad*, M. Kaufmann, A. Syeda, K. Ren, P. Abolmaesumi, J.F. Rudan, D. McKay, G. Fichtinger, P. Mousavi. ImSpect: Image-driven Self-supervised Learning for Surgical Margin Evaluation with Mass Spectrometry. International Journal of Computer Assisted Radiology and Surgery, special issue of International Conference on Information Processing in Computer-Assisted Interventions (IPCAI), 2024. 19(5), 1129–1136. (doi.org/10.1007/s11548-024-03106-1) (Shortlisted for Best Paper Award at IPCAI 2024).

J-9 D. Morton, **L. Connolly**, L. Groves, K. Sunderland, T. Ungi, A. Jamzad, M. Kaufmann, K. Ren, J. Rudan, G. Fichtinger, P. Mousavi. Development of a research testbed for intraoperative optical spectroscopy tumor margin assessment. Acta Polytechnica Hungarica - Special Issue on Digital Health Technologies, 2023 (acta-uni.obuda).

J-10 D. Liu, G. Li, S. Wang, Z. Liu, Y. Wang, **L. Connolly**, G. Shen, K. Cleary, I. Iordachita. An MR Conditional Robot for Lumbar Spinal Injection: Development and Preliminary Validation. The International Journal of Medical Robotics and Computer Assisted Surgery, 2023 (doi.org/10.1002/rcs.2618).

J-11 F. Fooladgar*, A. Jamzad*, **L. Connolly***, A. Santilli, M. Kaufmann, K. Ren, P. Abolmaesumi, J.F. Rudan, D. McKay, G. Fichtinger, P. Mousavi. Uncertainty estimation for margin detection in cancer surgery using mass spectrometry. International Journal of Computer Assisted Radiology and Surgery, 2022. 17(12):2305-13. (doi.org/10.1007/s11548-022-02764-3) * denotes equal contribution

J-12 **L. Connolly**, A. Deguet, S. Leonard, J. Tokuda, T. Ungi, A. Krieger, P. Kazanzides, P. Mousavi, G. Fichtinger, R.H. Taylor. Bridging 3D Slicer and ROS2 for image-guided robotic interventions. J. Sensors, 2022. 14 (5335). (doi.org/10.3390/s22145336) (Highlighted cover-page article)

J-13 **L. Connolly**, A. Jamzad, M. Kaufmann, C.E. Farquharson, K. Ren, J.F. Rudan, G. Fichtinger, P. Mousavi. Combined mass spectrometry and histopathology imaging for perioperative tissue assessment in cancer surgery. J. Imaging, 2021. 7(10), 203. (https://doi.org/10.3390/jimaging7100203) (Selected out of 27 articles as the cover story for journal issue: www.mdpi.com/2313-433X/7/10)

J-14 A. Santilli, A. Jamzad, N.N.Y Janssen, M. Kaufmann, **L. Connolly**, K. Vanderbeck, A. Wang, D. McKay, J.F. Rudan, G. Fichtinger, P. Mousavi. Perioperative Margin Detection In Basal Cell Carcinoma Using A Deep Learning Framework: A Feasibility Study. International Journal of Computer Assisted Radiology and Surgery, special issue of International Conference on Information Processing in Computer-Assisted Interventions (IPCAI), 2020. 15 (5), 887–896. (doi.org/10.1007/s11548-020-02152-9)

Full Conference Papers

C-1 S. Logan, **L. Connolly**, C. Barr, P. Mousavi, G. Fichtinger and K. Hashtrudi-Zaad. Evaluating Vibrotactile Feedback for Electromagnetic Surgical Navigation. SPIE Medical Imaging, 2026. (Submitted)

C-2 **L. Connolly**, A. Deguet, A. Munawar, T. Ungi, C. Yeung, R.H. Taylor, P. Mousavi, G. Fichtinger and K. Hashtrudi-Zaad. Fine-Tuning a Virtual Fixture Guidance System for Breast-Conserving Surgery based on Noticeable Differences. Hamlyn Symposium on Medical Robotics (HSMR), 2025. London, UK. (Proceedings: HSMR25-Proceedings-Final.pdf)

C-3 K. Hashtrudi-Zaad, C. Farvolden, **L. Connolly**, C. Barr, and G. Fichtinger. Robotic tracking of a resection cavity using a low cost bench-top robotic arm and electromagnetics. Medical Imaging 2025: Image-Guided Procedures, Robotic Interventions, and Modeling. Vol. 13408. SPIE, 2025. San Diego, USA. (doi.org/10.1117/12.3047143)

C-4 C. Farvolden, K. Hashtrudi-Zaad, **L. Connolly**, C. Barr, G. Fichtinger. An accessible 6-axis testbed for image-guided robotics research. Medical Imaging 2025: Image-Guided Procedures, Robotic Interventions, and Modeling. Vol. 13408. SPIE, 2025. San Diego, USA. (doi.org/10.1117/12.3047507) (3rd Place JMI Pitch Competition)

C-5 D. Liu, Z. Liu, G. Li, **L. Connolly**, G. Shen, and I. Iordachita. Development of an MRI Guided Auxiliary Robot for Spinal Injections. 2023 Seventh IEEE International Conference on Robotic Computing (IRC) (doi.org/10.1109/IRC59093.2023.00066).

C-6 A. Jamzad, F. Fooladgar, **L. Connolly**, D. Srikanthan, A. Syeda, K. Ren, S. Merchant, J. Engel, S. Varma, J.F. Rudan, G. Fichtinger, P. Mousavi. Bridging Ex-Vivo Training and Intra-operative Deployment for Surgical Margin Assessment with Evidential Graph Transformer. 2023 International Conference on Medical Image Computing and Computer-Assisted Intervention (pp. 562-571). Cham: Springer Nature Switzerland (doi.org/10.1007/978-3-031-43990-253)

C-7 **L. Connolly**, A. Deguet, T. Ungi, A. Kumar, A. Lasso, K. Sunderland, P. Kazanzides, A. Krieger, J. Tokuda, S. Leonard, G. Fichtinger, P. Mousavi, R.H. Taylor. An application of SlicerROS2: Haptic latency evaluation for virtual fixture guidance in breast conserving surgery. Hamlyn Symposium on Medical Robotics (HSMR), 2023. London, UK (Proceedings: HSMR23-Proceedings-Final.pdf)

C-8 O. Radcliffe, **Laura Connolly**, T. Ungi, C. Yeo, J.F. Rudan, G. Fichtinger, P. Mousavi. Navigated surgical resection cavity inspection for breast conserving surgery. Medical Imaging 2023: Image-Guided Procedures, Robotic Interventions, and Modeling. Vol. 12466. SPIE, 2023. San Diego, USA. (doi.org/10.1117/12.2654015)

C-9 D. Morton, **L. Connolly**, L. Groves, K. Sunderland, A. Jamzad, J.F. Rudan, G. Fichtinger, T. Ungi, P. Mousavi. Tracked tissue sensing for tumor bed inspection. Medical Imaging 2023: Image-Guided Procedures, Robotic Interventions, and Modeling. Vol. 12466. SPIE, 2023. San Diego, USA. (doi.org/10.1117/12.2654217)

C-10 **L. Connolly**, A. Jamzad, A. Nikniazi, R. Poushimin, J.M. Nunzi, J.F. Rudan, G. Fichtinger, P. Mousavi. Feasibility of combined optical and acoustic imaging for surgical cavity scanning." Medical Imaging 2022: Image-Guided Procedures, Robotic Interventions, and Modeling. Vol. 12034. SPIE, 2022. San Diego, USA. (doi.org/10.1117/12.2611964)

C-11 A. Greisman, A. Ivimey, **L. Connolly**, K. Hashtrudi-Zaad. Power based gravity compensation for flexible joint manipulators. IEEE International Symposium on Measurement and Control in Robotics (ISMCR), 2022. (doi.org/10.1109/ISMCR56534.2022.9950579)

C-12 **L. Connolly**, A. Deguet, K. Sunderland, A. Lasso, T. Ungi, J.F. Rudan, R.H. Taylor, P. Mousavi, G. Fichtinger. An Open-Source Platform for Cooperative Semi-Autonomous Robotic Surgery. 2021 IEEE International Conference on Autonomous Systems (ICAS), pp. 1–5, Aug. 2021. (doi.org/10.1109/ICAS49788.2021.9551149) (Best Student Paper Award)

C-13 F. Akbarifar, A. Jamzad, A. Santilli, M. Kaufmann, N. Janssen, **L. Connolly**, K. Ren, K. Vanderbeck, A. Wang, D. McKay, J.F. Rudan, G. Fichtinger, P. Mousavi. Graph-based analysis of mass spectrometry data for tissue characterization with application in basal cell carcinoma surgery. Medical Imaging 2021: Image-Guided Procedures, Robotic Interventions, and Modeling. Vol. 11598. SPIE, 2021. (doi.org/10.1117/12.2582045)

C-14 **L. Connolly**, A. Jamzad, M. Kaufmann, R. Rubino, A. Sedghi, T. Ungi, M. Asselin, S. Yam, J.F. Rudan, C. Nicol, G. Fichtinger, P. Mousavi. Classification of tumor signatures from electrosurgical vapors using mass spectrometry and machine learning: a feasibility study." Medical Imaging 2020: Image-Guided Procedures, Robotic Interventions, and Modeling. Vol. 11315. SPIE, 2020. Houston, USA. (doi.org/10.1117/12.2549343)

C-15 L. Yates, **L. Connolly**, A. Jamzad, M. Asselin, R. Rubino, S. Yam, T. Ungi, A. Lasso, C. Nicol, P. Mousavi, G. Fichtinger. Robotic tissue scanning with biophotonic probe. Medical Imaging 2020: Image-Guided Procedures, Robotic Interventions, and Modeling. Vol. 11315. SPIE, 2020. Houston, USA. (doi.org/10.1117/12.2549635)

C-16 **L. Connolly**, T. Ungi, A. Lasso, T. Vaughan, M. Asselin, P. Mousavi, S. Yam, G. Fichtinger. Mechanically controlled spectroscopic imaging for tissue classification. Medical Imaging 2019: Image-Guided Procedures, Robotic Interventions, and Modeling. Vol. 10951. SPIE, 2019. San Diego, USA. (doi.org/10.1117/12.2512481)

Conference Abstracts

A-1 **L. Connolly**, G. d'Albenzio, R. Hisey, T. Ungi, F. Saleck, P. Mousavi, R. Kikinis, Y. Ould Tfeil, A. Moulaye Idriss, G. Fichtinger. Print-and-Poke: A Low-Cost 3D-Printed Guidance Solution for Percutaneous Nephrostomy in LMICs. Hamlyn Surgical Robotics Challenge. (Best Application Award)

A-2 M. Bernardes, P. Moreira, **L. Connolly**, A. Deguet, S. Leonard, J. Tokuda. Integration of a ROS2-Based Robotic System with MRI-Guided Percutaneous Intervention Workflow. 36th Annual International Society for Medical Innovation and Technology Conference (iSMIT 2025), 2025, Bethesda North Marriott Hotel & Conference Center, Washington, DC, USA. (Accepted)

A-3 G. d'Albenzio, R. Hisey, **L. Connolly**, T. Ungi, F. Saleck, P. Mousavi, R. Kikinis, Y. Ould Tfeil, A. Moulaye Idriss, G. Fichtinger. Point-of-care Ultrasound Nephrostomy: A Practical Guidance System Tailored for Mauritania. 36th Annual International Society for Medical Innovation and Technology Conference (iSMIT 2025), 2025, Bethesda North Marriott Hotel & Conference Center, Washington, DC, USA. (Accepted)

A-4 J. Tokuda, P. Moreira, M. Bernardes, **L. Connolly**, A. Kumar, L. Al-Zogbi, L. Foley, K. Tuncali, C. Tempany, N. Hata, I. Iordachita, A. Krieger, A. Deguet, S. Leonard. Open-Source 3D-Printable MRI-Conditional Needle Placement Robot for Prostate Interventions. International Society for Magnetic Resonance in Medicine, Annual Meeting, 2025.

A-5 E. Elkind, O. Radcliffe, A. Tin Tun, **L. Connolly**, C. Davison, E. Purkey, G. Fichtinger, K. Thornton. Strengthening Low-cost Prosthetic Solutions in Thailand/Myanmar Through Academic Institution-NGO Collaboration. Health and Human Rights Conference, Queen's University, 2025. (3rd Place Research Competition)

A-6 **L. Connolly**, T. Ungi, A. Munawar, A. Deguet, C. Yeung, R.H. Taylor, P. Mousavi, G. Fichtinger, K. Hashtrudi-Zaad. A feasibility study on enhanced navigation in breast-conserving surgery through haptic feedback. The 2025 Image Guided Therapy x Imaging Network of Ontario Joint Symposium. (Best Presentation Award)

A-7 K. Hashtrudi-Zaad, C. Farvolden, **L. Connolly**, C. Barr, G. Fichtinger. Resection cavity tracking using a bench-top robot and electromagnetic tracking. The 2025 Image Guided Therapy x Imaging Network of Ontario Joint Symposium.

A-8 C. Farvolden, K. Hashtrudi-Zaad, **L. Connolly**, C. Barr, G. Fichtinger. Designing a 6-Axis Testbed for Accessible Image-Guided Robotics Research. The 2025 Image Guided Therapy x Imaging Network of Ontario Joint Symposium.

A-9 T. Elliott, F. Fooladgar, A. Jamzad, **Laura Connolly**, M. Kaufmann, K. Ren, J. Rudan, D. McKay, G. Fichtinger, and P. Mousavi. A Comparison of Uncertainty Techniques on Basal Cell Carcinoma Mass Spectrometry Data. The 2025 Image Guided Therapy x Imaging Network of Ontario Joint Symposium.

A-10 E. Elkind, A. Tin Tun, O. Radcliffe, **L. Connolly** , C. Davison, E. Purkey, P. Mousavi, G. Fichtinger, K. Thornton. Developing low-cost 3D-printed prosthetics with a functional wrist for patients along the Thai-Myanmar border. The 2025 Image Guided Therapy x Imaging Network of Ontario Joint Symposium.

A-11 M. Farahmand, A. Jamzad, F. Fooladgar, **L. Connolly**, M. Kaufmann, K. Yi Mi Ren, J.F. Rudan, D. McKay, G. Fichtinger, P. Mousavi. Foundation Models for Cancer Tissue Margin Assessment with Mass Spectrometry. The 2025 Image Guided Therapy x Imaging Network of Ontario Joint Symposium.

A-12 E. Elkind, A. Tin Tun, O. Radcliffe, **L. Connolly**, C. Davison, E. Purkey, P. Mousavi, G. Fichtinger, and K. Thornton. Enhancing healthcare access by developing low-cost 3D printed prosthetics along the Thai-Myanmar border. The Canadian Conference on Global Health, 2024.

A-13 **L. Connolly**, A. Jamzad, M. Kaufmann, D. McKay, G. Fichtinger, P. Mousavi. Real-time margins for skin cancer excision using mass spectrometry analysis of cautery smoke. The 77th Annual Meeting of the Canadian Society of Plastic Surgeons, 2024.

A-14 **L. Connolly**, H. Song, K. Xu, A. Deguet, S. Leonard, G. Fichtinger, P. Mousavi, R.H. Taylor, E. Boctor. Photoacoustic detection of residual cancer in breast-conserving surgery. The 22nd Annual Symposium of the Imaging Network of Ontario (ImNO), 2024. (Best Presentation Award)

A-15 O. Radcliffe, A. Tin Tun, N. Chan, **L. Connolly**, T. Ungi, K. Thornton, C. Davison, E. Purkey, P. Mousavi, G. Fichtinger. 3D printing prosthetics on the Thailand-Myanmar border. The 22nd Annual Symposium of the Imaging Network of Ontario (ImNO), 2024.

A-16 **L. Connolly**, A. Deguet, T. Ungi, A. Kumar, A. Lasso, K. Sunderland, P. Kazanzides, A. Krieger, J. Tokuda, S. Leonard, G. Fichtinger, P. Mousavi, R.H. Taylor. Haptic feedback and ultrasound guidance for lumpectomy with SlicerROS2. ASMUS workshop at MICCAI, 2023. (Best Demo Award)

A-17 **L. Connolly**, A. Jamzad, F. Fooladgar, A. Syeda, M. Kaufmann, K. Ren, P. Abolmaesumi, J.F. Rudan, D. McKay, G. Fichtinger, P. Mousavi. Margin detection in skin cancer surgery via 2D representation of mass spectrometry data. The 21st Annual Symposium of the Imaging Network of Ontario (ImNO), 2023.

A-18 D. Morton, **L. Connolly**, L. Groves, K. Sunderland, A. Jamzad, J.F. Rudan, G. Fichtinger, T. Ungi, P. Mousavi. Evaluation of tracked optical tissue sensing for tumor bed inspection. The 21st Annual Symposium of the Imaging Network of Ontario (ImNO), 2023.

A-19 O. Radcliffe, **L. Connolly**, T. Ungi, C. Yeo, J.F. Rudan, G. Fichtinger, P. Mousavi. Electromagnetic navigation for residual tumor localization in breast-conserving surgery. The 21st Annual Symposium of the Imaging Network of Ontario (ImNO), 2023.

A-20 A. Jamzad, **L. Connolly**, F. Fooladgar, M. Kaufmann, K. Ren, S. Merchant, J. Engel, S. Varma, P. Abolmaesumi, G. Fichtinger, J.F. Rudan, P. Mousavi. Deployment of deep models for intra-operative margin assessment using mass spectrometry. Neural Information Processing Systems (MED-NEURIPS), 2022.

A-21 **L. Connolly**, A. Jamzad, A. Nikniazi, R. Poushimin, A. Lasso, K. Sunderland, T. Ungi, J. Michel Nunzi, J.F. Rudan, G. Fichtinger, P. Mousavi. An open-source testbed for developing image-guided robotic tumor-bed inspection. The 20th Annual Symposium of the Imaging Network of Ontario (ImNO), 2022.

A-22 **L. Connolly**, K. Sunderland, A. Deguet, A. Lasso, T. Ungi, J.F. Rudan, R.H. Taylor, P. Mousavi, G. Fichtinger. A platform for robot-assisted intraoperative imaging in breast conserving surgery. The 19th Annual Symposium of the Imaging Network of Ontario (ImNO), 2021. (Best Pitch Award)

A-23 **L. Connolly**, A. Jamzad, M. Kaufmann, R. Rubino, A. Sedghi, T. Ungi, M. Asselin, S. Yam, J.F. Rudan, C. Nicol, G. Fichtinger, P. Mousavi. Classification of primary tumor and surrounding tissue in breast cancer xenograft models. The 18th Annual Symposium of the Imaging Network of Ontario (ImNO), 2020.